

KG-RAG

Literature Review

Knowledge Graph-Augmented Retrieval for Large Language Models

MULTI-HOP QA · MEDICAL NLP · BIOMEDICAL KG

Papers Covered

- 01 **SG-RAG**
ICNLS 2024
- 02 **HyKGE**
ACL 2025
- 03 **KRAGEN**
Bioinformatics 2024
- 04 **KG²RAG**
NAACL 2025

Why Standard RAG Falls Short



Hallucination

LLMs generate false facts for niche domains where training data is sparse or unreliable.



Multi-hop Failure

Retrieval usually does one semantic lookup, missing intermediate steps ($A \rightarrow B \rightarrow C$) required for reasoning.



Noisy Retrieval

Semantic similarity often returns homogenous chunks that lack diversity or logical connectivity.



Solution: Use Knowledge Graph structure to guide, expand, and validate retrieval paths.

Four Papers at a Glance

Paper	Venue	Core Problem	Key Innovation	Domain
SG-RAG	ICNLSP '24	Multi-hop QA Fails	Cypher Subgraph Retrieval	Movies
HyKGE	ACL '25	Vague Medical Queries	Hypothesis Expansion	Medical
KRAGEN	Bioinfo '24	Complex Biomedical QA	Graph-of-Thoughts	Biomedical
KG ² RAG	NAACL '25	Isolated Chunks	MST Paragraph Org.	Open QA

Paper 01

SG-RAG

SubGraph Retrieval Augmented Generation for Multi-Hop QA

Neo4j KG

Zero-Shot

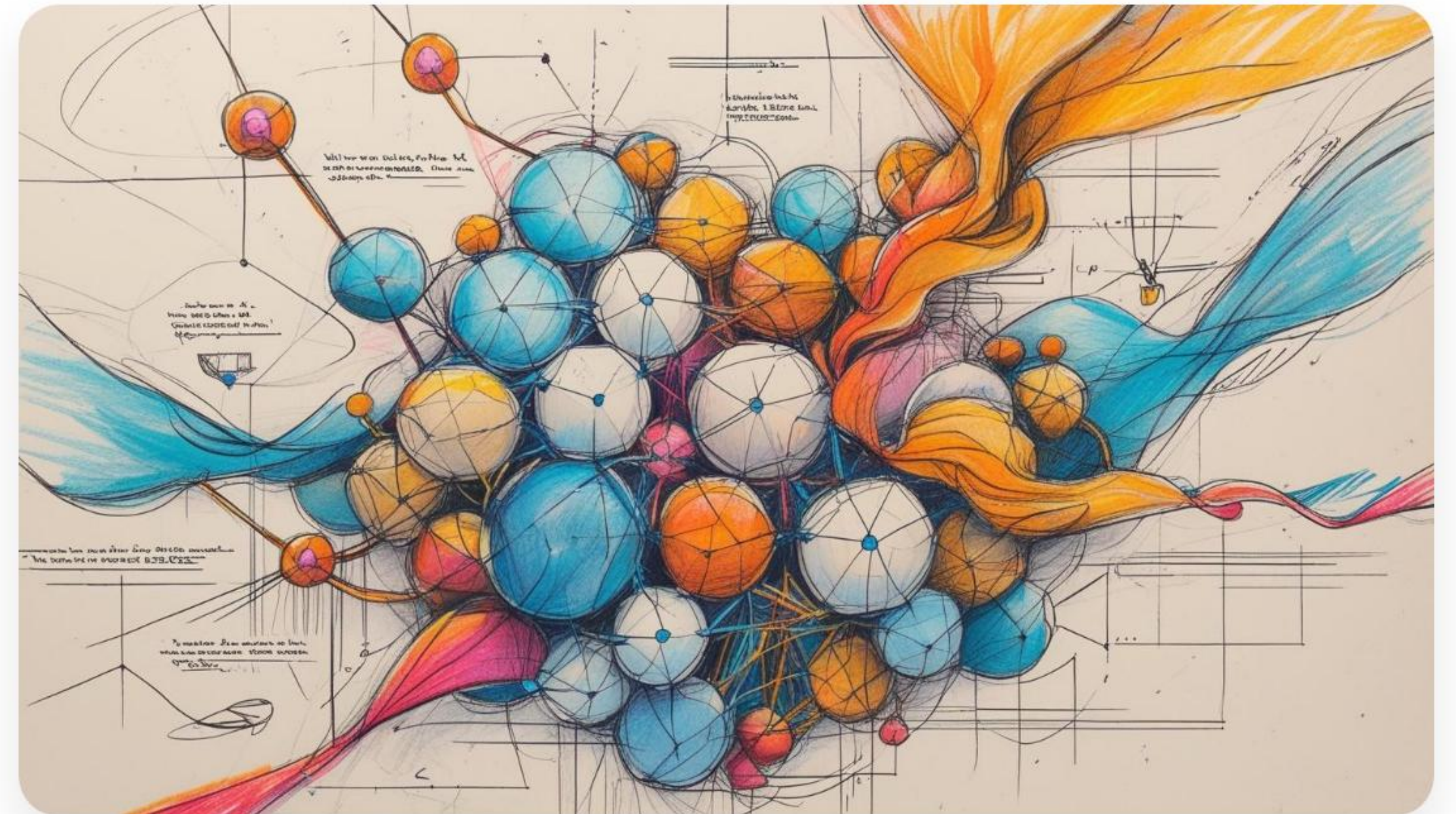
Multi-Hop Reasoning Gaps

The "Director" Challenge

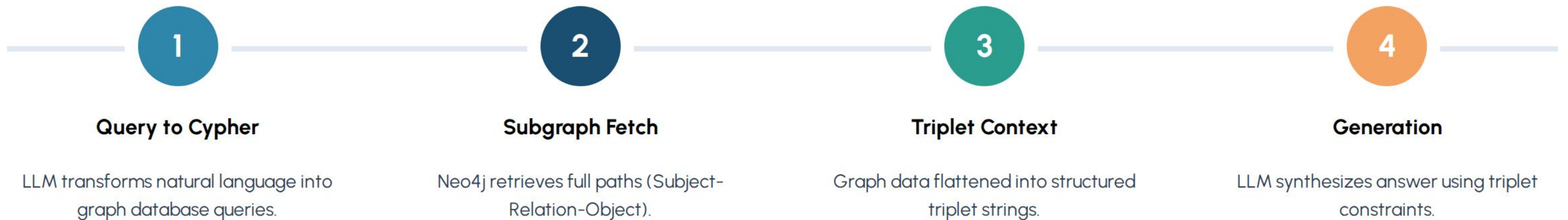
Standard RAG fails when the query doesn't explicitly contain the intermediate entity.

Query: "Years films led by director of Edges of the Lord were released?"

1. Direct Link: Edges of the Lord → Director X
2. Expansion: Director X → Film Y, Film Z
3. Target: Film Y → 1993, Film Z → 1987

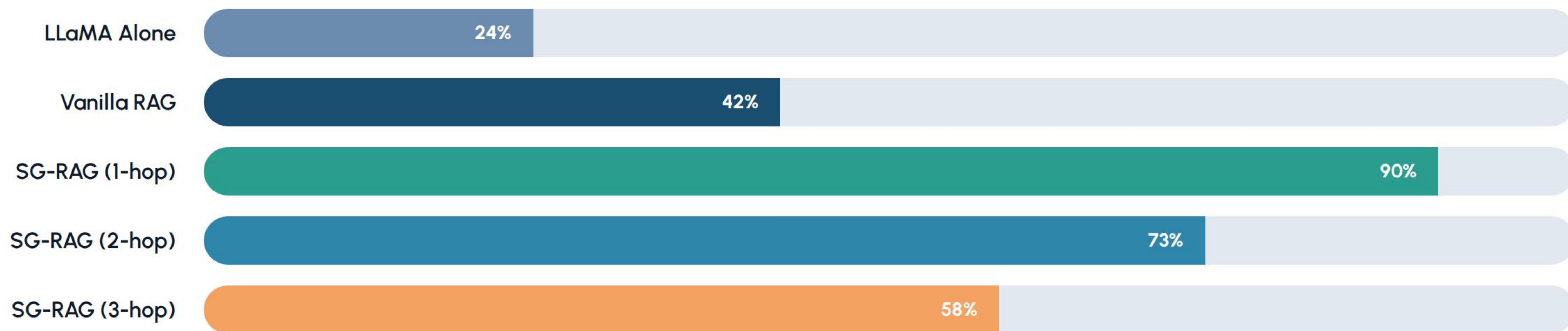


SG-RAG Pipeline



```
MATCH (a:People)-[r1:DIRECTED]→(m:Movie)-[r2:RELEASE_YEAR]→(y:Year)
WHERE a.name="Director X" RETURN a, r1, m, r2, y
```

Performance on MetaQA



SG-RAG maintains significantly higher precision as hop complexity increases compared to baseline models.

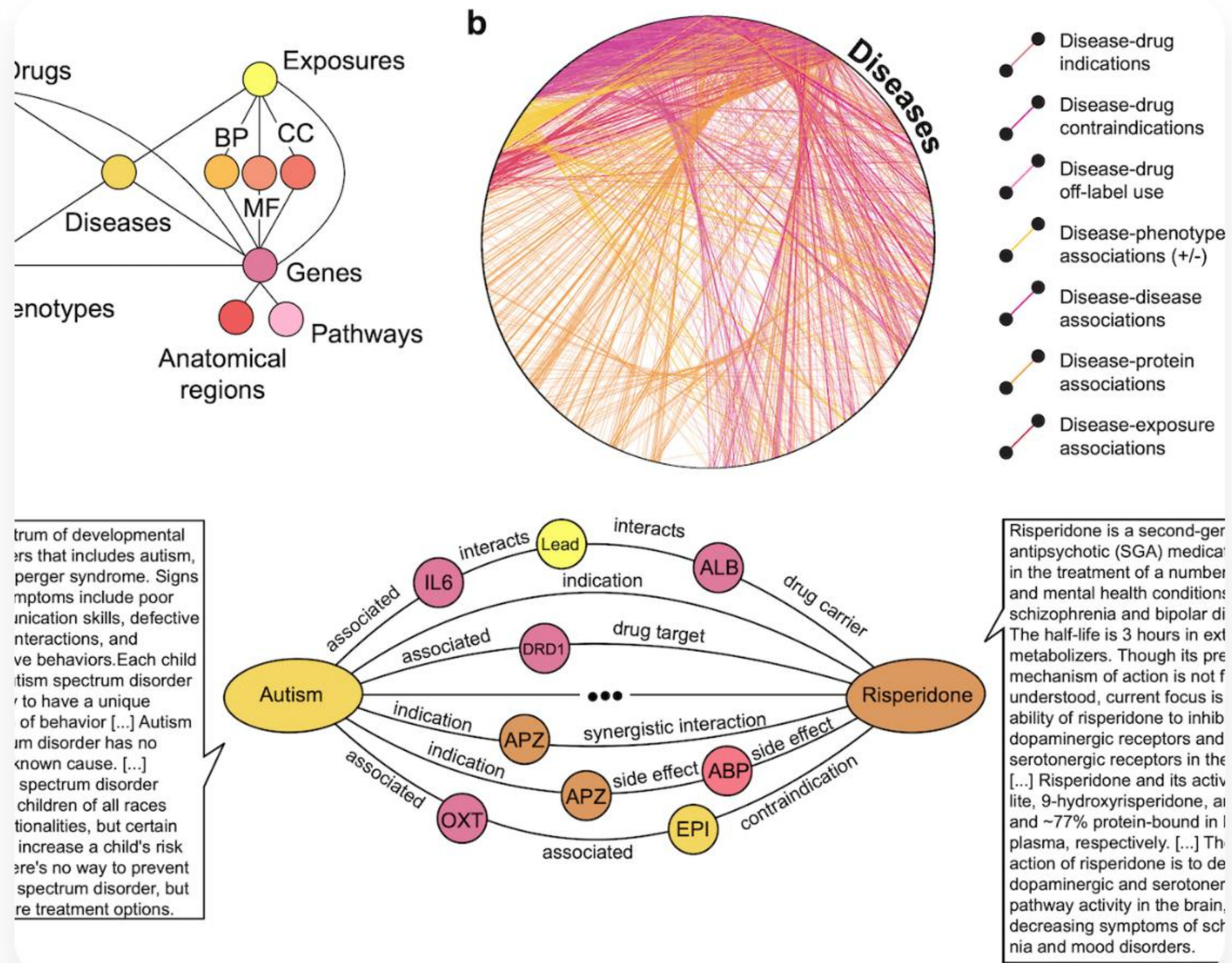
Paper 02

HyKGE



Hypothesis Knowledge Graph Enhanced RAG for Medical LLMs

Medical Reasoning Challenges



Vague Queries

Patients describe symptoms, not technical medical terms, making direct graph lookup difficult.

LLM Overhead

Iterative graph traversal is too slow for clinical use cases (4-5 calls per query).

Noisy KG Chains

Medical KGs are dense; standard ranking returns repetitive results.

HyKGE: Efficient Clinical QA

+28%

ACCURACY ON MEDICAL
BENCHMARKS

2x

FASTER (ONLY 2 LLM
CALLS)

+84%

CONTEXT RELEVANCE
SCORE

-57%

REDUCTION IN
HALLUCINATION



Innovation: Hypothesis Expansion

LLM generates a "potential" answer first. HyKGE uses these terms (symptoms, drugs, mechanisms) to enrich the search anchor in the KG before the final response generation.

Paper 03

KRAGEN

Knowledge Retrieval Augmented Generation Enhanced Network

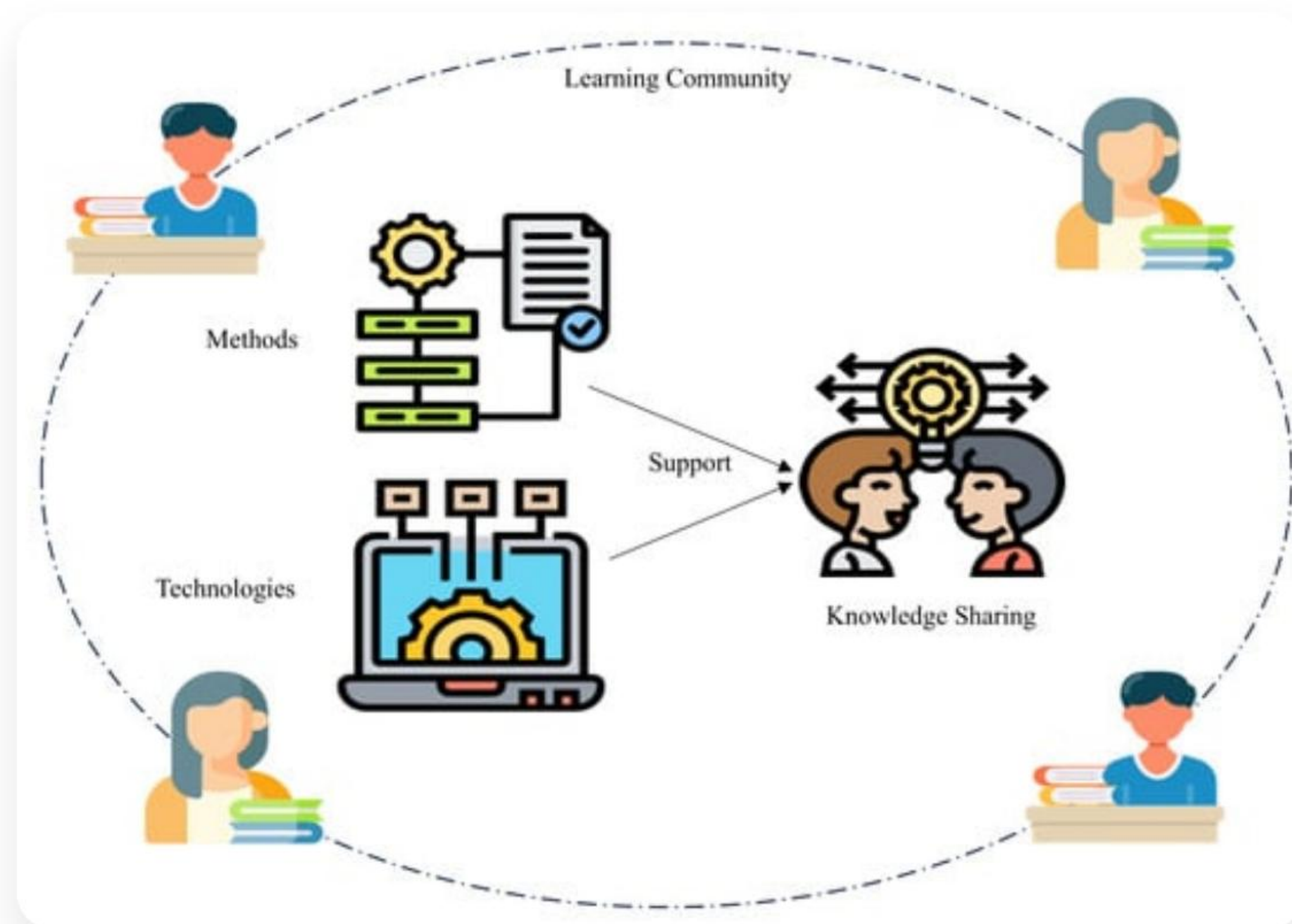
Biomedical Graph-of-Thoughts

Decomposing Complexity

Answering multi-gene biomedical questions requires breaking down the problem into logical vertices.

- **Vertex A:** Gene-Protein mapping
- **Vertex B:** Disease-pathway association
- **Edges:** Flow of derived context between reasoning steps

Tech Stack: Weaviate DB + hybrid search (dense similarity + keyword matching).

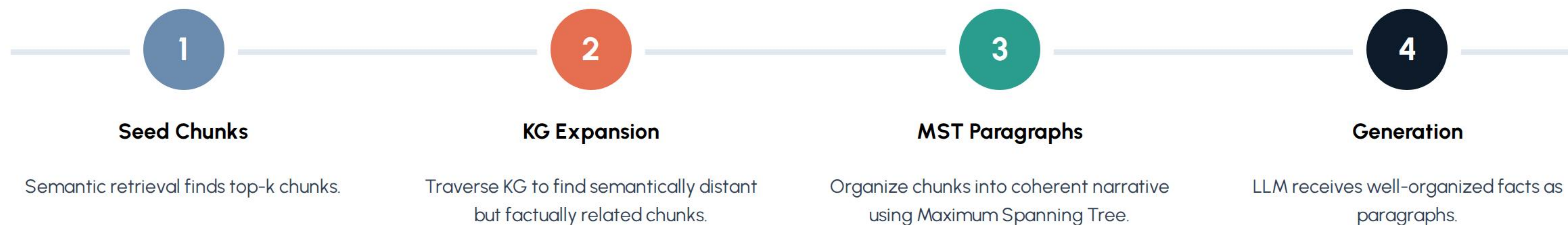


Paper 04

KG²RAG

Knowledge Graph-Guided Retrieval Augmented Generation

Beyond Isolated Chunks



Key Insight: Standard RAG retrieves fragmented text. KG²RAG uses graph connectivity to "stitch" isolated document pieces into a logically flowing paragraph for the LLM.

Pipeline Intervention Points

Phase	SG-RAG	HyKGE	KRAGEN	KG ² RAG
Pre-Retrieval	Cypher Generation	Hypothesis Expansion	GoT Decomposition	Standard Lookup
Retrieval	Subgraph Match	3-Chain Types	Vertex-based RAG	M-hop Expansion
Post-Retrieval	Triplet Context	Fragment Reranking	GoT Result Flow	MST Organization
Generation	Zero-Shot Triplet	Structured Prompt	Vertex Synthesis	Narrative Chunks

4 Shared Themes

Graph > Similarity

All papers agree: structured graph traversal beats semantic embedding similarity for complex, multi-step knowledge requirements.

Triplets as Context

Converting KG data to (Subject, Relation, Object) triplets provides directionality-aware context that raw text lacks.

Noise Management

Post-retrieval filtering is essential. Too much knowledge triggers the "lost-in-the-middle" effect and hurts accuracy.

Zero-Shot Portability

The most effective frameworks avoid expensive fine-tuning, focusing on prompt engineering and graph database queries for domain agility.

Future Research Gaps

Automatic Schema Alignment

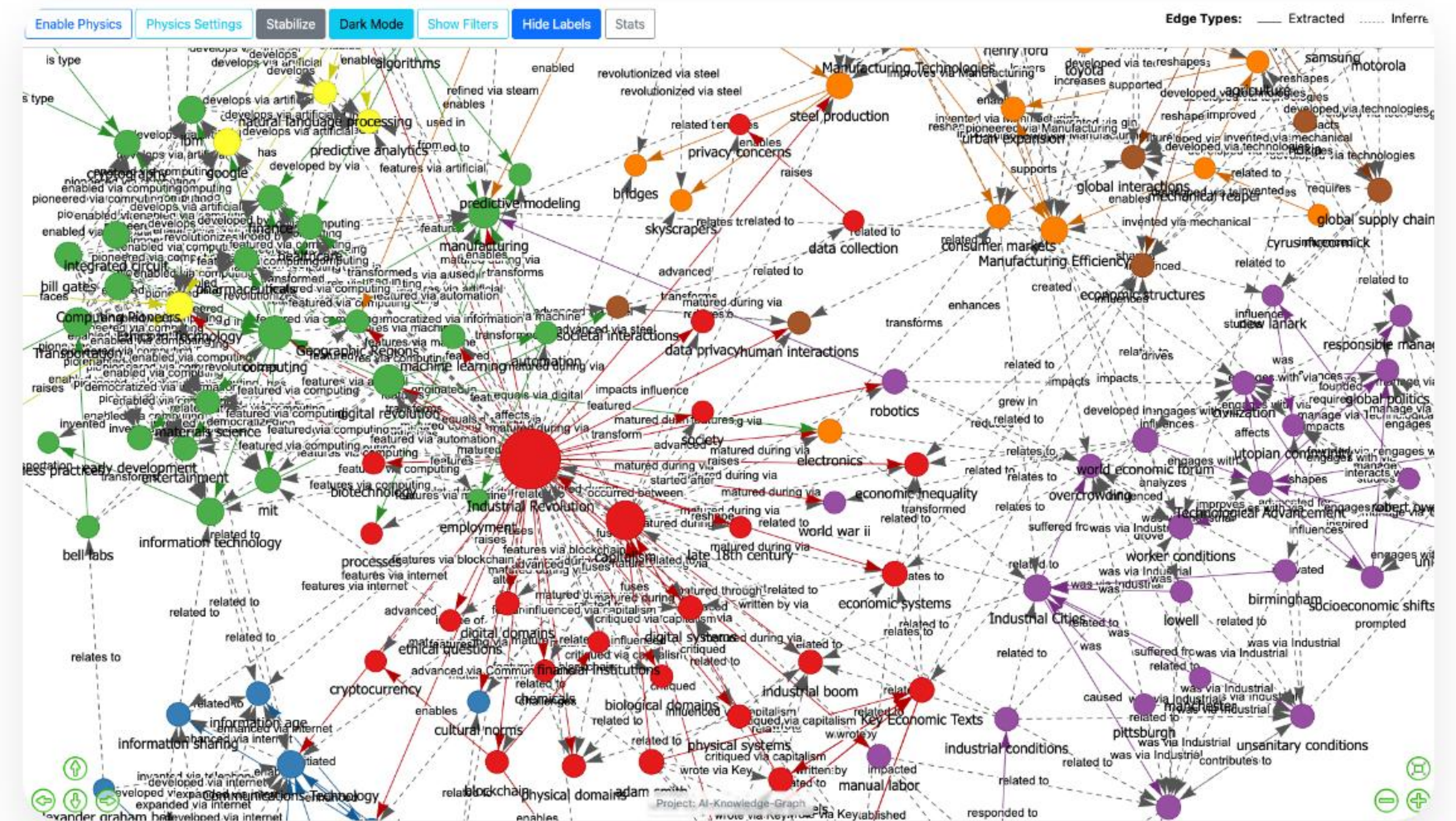
How to handle cross-domain KGs (e.g., medical → legal) without manual Cypher templates.

Entity Disambiguation

Robust linking for queries with typos, aliases, or colloquialisms in specialized domains.

Dynamic Graph Updates

Managing real-time KG expansion as new documents are added to the RAG collection.



Key Takeaways

1. Standard RAG is a "point search"; KG-RAG is a "path traversal" strategy essential for multi-hop reasoning.
2. Graph topology encodes logic that embeddings cannot capture—specifically causality and multi-entity relationships.
3. Hybrid approaches (MST paragraphs, GoT) bridge the gap between structured databases and unstructured LLM inputs.

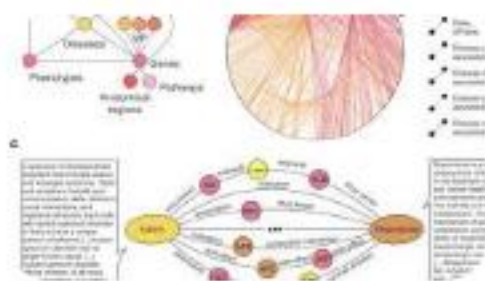
Questions? | contact@research-lab.ai

Image Sources



<https://helabenhalfallah.com/wp-content/uploads/2025/07/Graph-Neural-Networks-and-the-Shape-of-Thought-article-cover-AI-context.jpg>

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